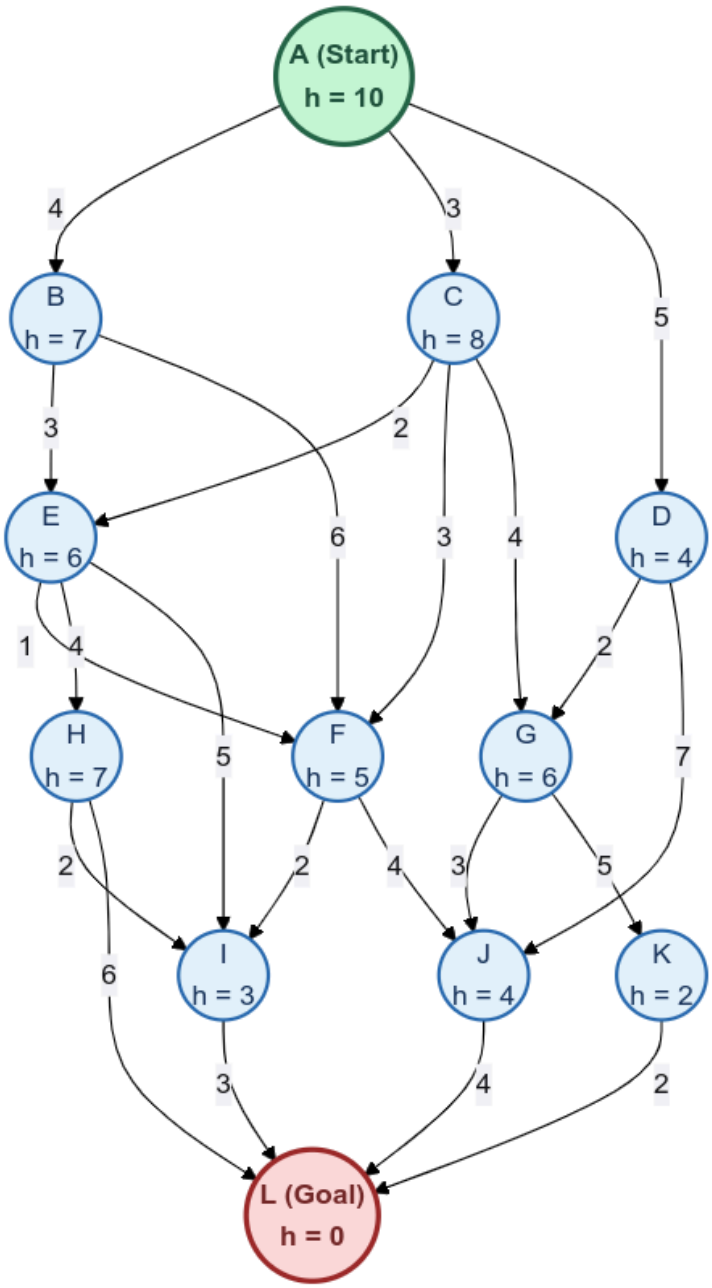
Question 1

Fill up the table below with actual cost taken from each Node to Goal L. Based on the costs calculated and the given heuristic value. answer the following questions

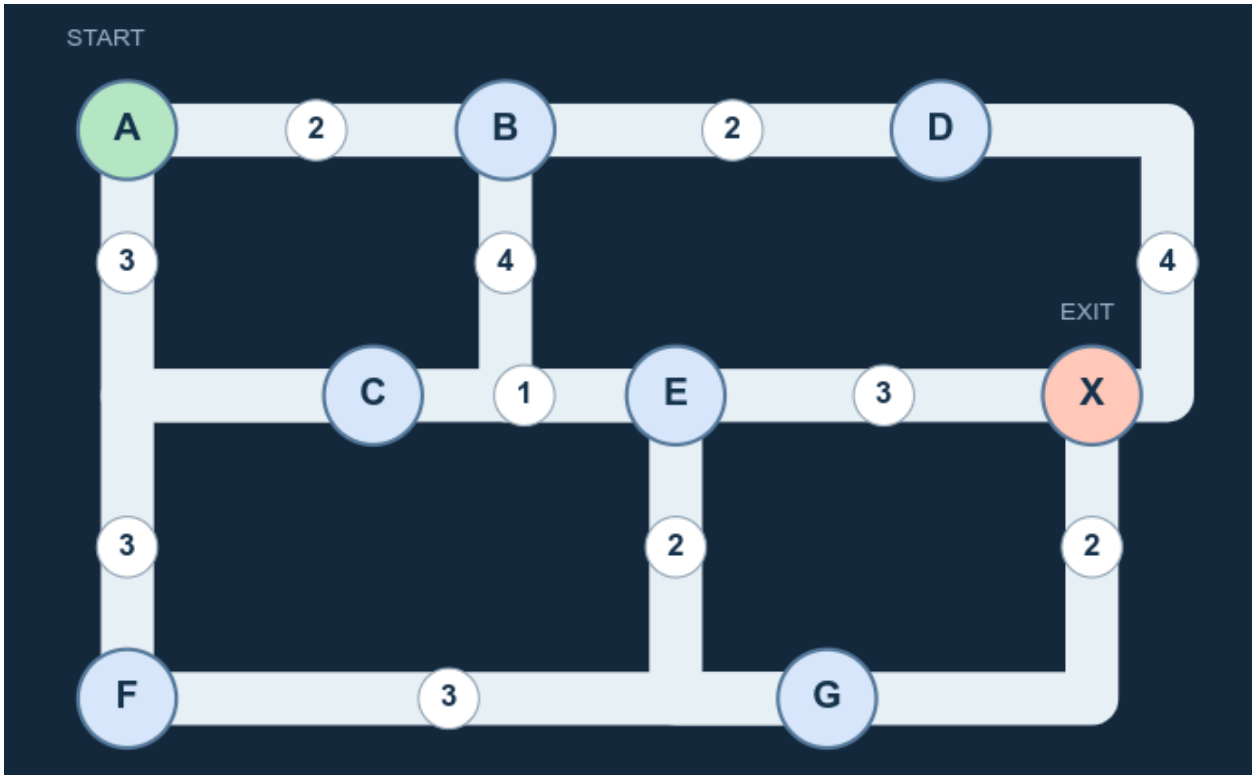
1. Considering A as the start node, apply Tree version of A* Search on the given graph to find the shortest path and the cost of the path.
2. Considering A as the start node, apply Tree version of Greedy Best First Search on the given graph to find the shortest path and the cost of the path
3. Is the given heuristic admissible? If not, Identify the nodes that violate admissibility.



Node	h(L)	Cost
A	10	
B	7	
C	8	
D	4	
E	6	
F	5	
G	6	
H	7	
I	3	
J	4	
K	2	
L	0	

Question 2

Start at A and reach the exit X. The pale corridors are the only legal routes. Each numbered badge gives the travel cost for that entire corridor. Corridors can be traveled in both directions.



h(A)	h(B)	h(C)	h(D)	h(E)	h(F)	h(G)	h(X)
7	5	6	4	3	4	2	0

1. Apply graph-search A* from A to X using f = g + h. Show the expansion order, path, and total cost.
2. Apply graph-search Greedy Best-First Search using h alone. Show the expansion order, path, and total cost.
3. Determine whether the heuristic is admissible. For each violating room, replace its estimate with the maximum admissible value.

Break ties alphabetically. Stop when X is removed from the frontier. For A*, update a room when a cheaper route is found and reopen expanded rooms if necessary. For Greedy, keep the first discovered route to each room.

Question 3

Showcase a graph of five nodes where the heuristic is admissible but not consistent.

Question 4

Consider a 2D Maze Problem, where you have to reach G, starting from S.

- The cost of horizontal/vertical movement is 1, and the cost of diagonal movement is 2.
- You can not move to the cells marked with X.
- Moving to a cell marked with 2, increases path cost by 2.

Define a suitable heuristic on your own and simulate graph A* search from start state S to Goal Node G.

Break ties alphabetically.

2					G
	X	X	X	2	
				X	
	2	2	X		
S			X		